Ben Phan

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EDUCATION

National Cheng Kung University (NCKU) *Sep 2023 –Aug 2025*
Master of Science in Computer Science
- Advisor: Prof. Jung-Hsien Chiang
- Vice President of the Vietnamese Student Association at NCKU (2024 - 2025).
- Outstanding Thesis Award, invited to meet the Vice President at the Presidential Office of Taiwan.

Danang University of Science & Technology *Sep 2018 –Mar 2023*
Bachelor of Engineering in Control Engineering and Automation
- Advisor: Ph.D. NGO Dinh-Thanh
- Class President (2019 - 2023).
- Vice President of the BK Maker Scientific Research Club (2021 - 2022).

RESEARCH INTERESTS

My research interests include Artificial Intelligence, Machine Learning, and Deep Learning, with a focus on Natural Language Processing, Computer Vision, and AI Agents. I work on optimizing Large Language Models (LLMs) and deep learning architectures for efficient deployment on edge and local devices, exploring techniques such as model compression, retrieval-augmented generation, and knowledge graph integration to enhance reasoning, scalability, and performance under resource constraints.

PUBLICATIONS AND SUBMISSIONS

MyMy at SemEval-2025 task 9: A robust knowledge-augmented data approach for reliable food hazard detection
Ben Phan, Jung-Hsien Chiang
Proceedings of the 19th International Workshop on Semantic Evaluation, an ACL Workshop, 2025
Highest average performance at SemEval-2025 Task 9, ACL Workshop, Vienna, Austria.

CAS: enhancing implicit constrained data augmentation with semantic enrichment for biomedical relation extraction and beyond
Fang-Yi Su, Gia-Han Ngo, **Ben Phan**, Jung-Hsien Chiang
Journal of Database (Q1), 2025

Optimized biomedical entity relation extraction method with data augmentation and classification using GPT-4 and Gemini
Cong-Phuoc Phan, **Ben Phan**, Jung-Hsien Chiang
Journal of Database (Q1), 2024

Probability Model with Ensemble Learning and Data Augmentation for Named Entity Recognition (NER) and Relation Extraction (RE) Tasks
Cong-Phuoc Phan, Gia-Han Ngo, **Ben Phan**, and Jung-Hsien Chiang
Proceedings of the BioCreative VIII: Curation and Evaluation in the era of Generative Models, 2023
Ranked among the Top 3 overall in the BioCreative VIII Challenge 2023.

Deep learning framework applied for predicting anomaly of respiratory sounds
Dat Ngo, Lam Pham, Anh Nguyen, **Ben Phan**, Khoa Tran, Truong Nguyen
International Symposium on Electrical and Electronics Engineering (ISEE), 2021

EXPERIENCE

Research Development Software Engineer — *Hsinchu, Taiwan*
群聯電子 *PHISON Electronics* — *Sep 2025 –Now*

- Develop firmware for SSD controller, focusing on Flash Physical Layer (FPL), error handling, and decoding flow.
- Research AI agents and LLM-based approaches for firmware development, aiming to improve automation, debugging, and system optimization.

AI Researcher — *Tainan, Taiwan*
Intelligent Information Retrieval Lab (IIR Lab), CSIE, NCKU — *Sep 2023 –Aug 2025*

- Developing and refining RAG techniques for enhanced information retrieval.
- Investigating Graph-based RAG models to improve the accuracy and efficiency of retrieval.
- Exploring integration of Large Language Models with retrieval systems to enhance the quality of generated information.

AI Engineer — *Hanoi, Vietnam*
Vingroup Big Data Institute — *Jun 2022 –Sep 2023*

- Completed courses in Linear Algebra, Probability & Statistics, Machine Learning, Deep Learning, Generative Model, Computer Vision, Natural Language Processing, and AI Ethics.
- Developed Face ID, Car Parking, and Uncovered Truck detection for VinFast factories and VinHome, processing data from 2,000+ cameras using GStreamer, DeepStream, TensorRT, and Triton.
- Optimized and deployed computer vision models (QAT, PQT) for VinFast Smart Factory, improving AI performance in industrial automation.

AI Engineer Internship — *Danang, Vietnam*
FTECH CO., LTD — *May 2021 –Jun 2022*

- Gained foundational knowledge of AI, deep learning, and computer vision techniques.
- Acquired hands-on experience with cloud platforms such as Azure and AWS for deploying AI models and services.

Embedded Software Internship — *Danang, Vietnam*
Da Nang Software Development Centre — *May 2019 –Jun 2020*

- Developed IoT-based tracking system for real-time bus movement monitoring in Danang.
- Gained proficiency in C, C++, and Linux environments through embedded software projects.

SKILLS

Computer Languages: Python, C++, C, Java, Bash, Linux.

Machine Learning Libraries: Pytorch, Tensorflow, Sklearn, Numpy, Triton, TensorRT, DeepStream, Gstreamer, Transformers, Langchain, LLMs, RAG.

Cloud & Tools: AWS & Azure, Docker, Git.

Languages: Vietnamese, English, Chinese.

ACHIEVEMENTS

- Outstanding Thesis Award (Top 1%) 18th TSC AI Application Competition. *2025*
- NCKU Distinguished International Student Scholarship for Master's program. *2023 –2025*
- Highest average performance at SemEval-2025 Task 9, ACL Workshop, Vienna, Austria. *2025*
- Ranked among the Top 3 overall in the BioCreative VIII Challenge. *2024*
- Academic Excellence Scholarship (Top 5%) at UD - DUT for outstanding performance. *2023*
- Third Prize in USAID BUILD-IT & Dow Vietnam STEM. *2023*
- Vingroup AI Engineer Training Scholarship, selected for prestigious AI training program. *2022*
- Second Prize in Science Research, UD - DUT. *2022*
- Second Prize in BK-Techshow, UD - DUT. *2022*
- Top 8 in Siemens Vietnam SIMATIC Digital Transformation Contest. *2021*
- Appropriate Technology Award, University of Tsukuba, Japan. *2021*
- Top 6 & Creativity Award in ABU Robocon Asia-Pacific. *2020*
- Certificate of Merit in COVID-19 Community Service Project. *2020*
- Learning Express Program, Singapore Polytechnic Exchange. *2019*

REFERENCES

Prof. Jung-Hsien Chiang
Distinguished Professor, Department of Computer Science and Information Engineering, NCKU.

Ph.D. NGO Dinh-Thanh
Lecturer at The University of Danang, University of Science and Technology.